

When Reported Joint Angles Are Not Enough: Local Calibration of a Low-Cost Printed Arm

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Abstract—Forward kinematics usually treats the angles a robot’s servos report as the arm’s physical state. We tested that assumption on one printed SO-ARM101 with a wrist camera and a fixed ChArUco board. Readings repeated after the same command varied by less than a millimetre, but reversing the approach direction moved the mean position by 10 to 14 mm. Across a workspace survey, switching from commanded to reported angles cut the mean disagreement with the camera from about 16 to 4 mm, and a repeat survey seven weeks later gave the same numbers. A local fit of the chain halved what remained. A local model of the command-to-position map, fitted in one session, was tested on 20 held-out targets approached once without and once with it. Of the 13 measured in both passes, all 13 improved, and the mean distance from the aim point fell from 22 to 6 mm. The same kind of analytic fit left about 15 mm on a more deeply flexed set of poses, more than the nominal chain leaves there. The fitted parameters are best read as an operating condition rather than as the geometry of the arm.

Index Terms—robot calibration, low-cost manipulators, kinematic identification, external measurement, compliance, calibration transfer

I. INTRODUCTION

Robot software often estimates tool position by applying forward kinematics to the joint angles reported by the motors. The estimate is always available, which makes it convenient for labelling demonstrations, defining workspaces, and expressing actions in task space. It is still an estimate made from sensors inside the mechanism. Backlash, compliance, assembly error, and gravity can all move the end of the arm without producing a matching change in the reported angles.

That gap matters for the inexpensive printed arms now used in open robot-learning stacks and compact vision-language-action models [1], [2]. A controller can settle in the same place every time while the task-space position inferred from its command is wrong by more than a centimetre, and the one inferred from its encoders by several millimetres. A calibration fitted on such an arm can be accurate on the poses used to fit it and still absorb deformation that belongs to those poses alone. The paper turns on one distinction: the command, the reported state, and the physical state are three different things, and the experiments below measure the gaps between them.

This paper answers three questions on one such arm, in order. Does a single command produce a single physical position? How much of the disagreement between kinematics and an external camera do the reported angles remove? And when a local model does correct the arm, do its parameters describe the machine or only the poses it was fitted on?

II. RELATED WORK

Robot calibration uses measured constraints or external position data to identify a kinematic model and compensate its error. Recent work has made the process cheaper [3], combined rigid kinematics with learned or elastic residual models [5], and treated calibration parameters as configuration dependent when the load changes across the workspace [4]. Backlash, thermal state, and payload are known to move the fitted command-to-position relation [6], [7]. Those results motivate the hybrid model used here. This paper offers no new calibration algorithm. It measures what encoder feedback and a local fit can account for on a low-cost printed arm, and what they leave behind. ISO 9283 remains the reference for separating pose accuracy from repeatability [8].

III. SETUP

The test arm was a five-joint SO-ARM101 with printed links and serial-bus servos. Its joints are named base rotation, shoulder pitch, elbow, wrist pitch, and wrist roll. A camera fixed to the wrist viewed a 7×5 ChArUco board taped to the table (Fig. 1).

A reading is the average of five standstill frames, each with at least ten detected corners. When the board was not visible the reading was recorded as missing rather than interpolated, which matters for several results below. Before the experiments we measured known spans on the board through the full measurement pipeline. They agreed with their true lengths to within a millimetre. That check constrains scale near the centre of the view and does not independently validate depth or board tilt.

Kinematics and camera must share one frame before they can be compared. A rigid board-to-base transform and a fixed camera offset in the wrist frame provide that alignment, and the link geometry stays nominal for the comparison in Section V. Each pose therefore yields three estimates of the same camera position: forward kinematics at the commanded angles, forward kinematics at the reported angles, and the camera itself.

The sign convention of each joint was checked against the camera by flipping the recorded sign of one joint at a time and refitting the alignment. The wrist roll, as recorded by the servo calibration, ran against the model’s axis: flipped, the alignment residual fell from about 22 to 4 mm, and no other flip helped. Every joint angle in this paper carries that correction. The last session, on 23 September, set the corrected sign in the servo calibration before any measurement, and the same check on

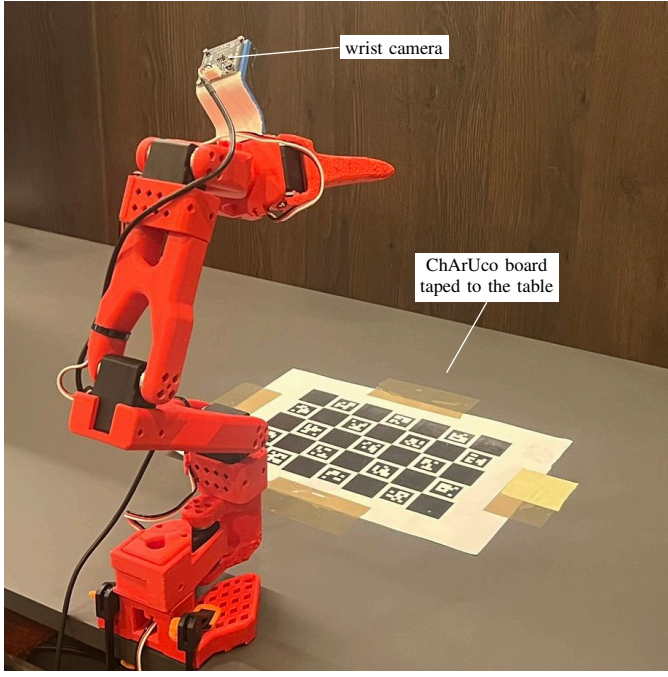


Fig. 1. Measurement setup: a camera on the wrist observes a ChArUco board taped to the table.

its runs found no flip that helped. The nominal geometry is the published SO-ARM100 chain.

IV. ONE COMMAND, TWO POSITIONS

The first question is whether a command names a position at all. We sent one command 50 times in each of five conditions: two approach directions while the arm was warm, one approach again after a brief power cycle while still warm, and the same two approaches the next morning, as the first motion after a night powered off. Servo temperatures were 37 to 43 °C in the warm blocks and 29 to 34 °C in the cold ones. The two approaches were interleaved inside each thermal block, and every reading was kept.

Under a fixed approach the readings clustered tightly: the pooled RMS distance of the readings from their own condition mean was 0.67 mm. Reversing the approach then moved the mean camera position by 14.3 mm while warm and by 10.0 mm cold. Thermal state and power cycling mattered far less. Under the same approach, the overnight cool-down moved the mean by 1.6 mm and the brief power cycle by 1.1 mm. Fig. 2 shows all four shifts. The arm is repeatable, but the command alone does not select one physical position. These readings do not separate backlash, compliance, and thermal effects from one another.

V. REPORTED ANGLES CLOSE MOST OF THE GAP

If the command is ambiguous, the reported angles are the obvious remedy. To see how much they help, we ran a workspace survey. It walked a fixed random sequence of candidate poses, each commanded from the same home position, until 25 had returned a reading. That took 33 attempts, and the

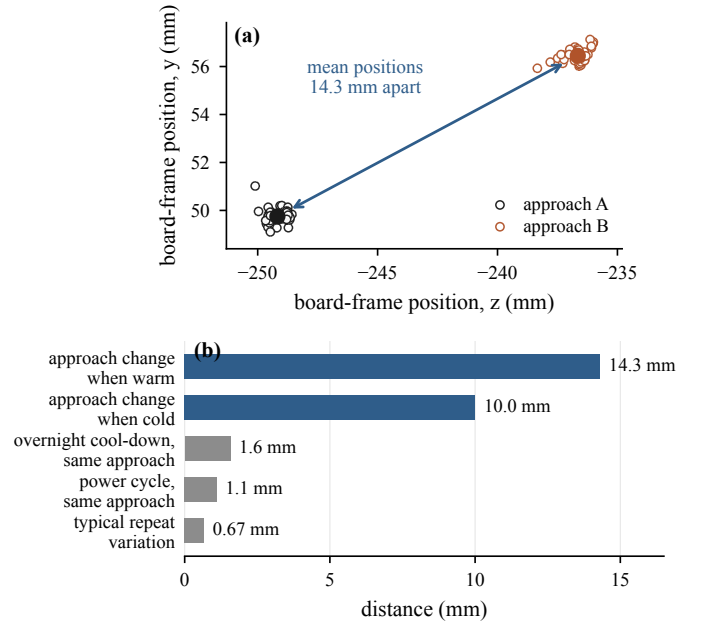


Fig. 2. Repeatability at a single command. (a) Each warm approach forms a tight cluster, and the two clusters do not share a centre. (b) Reversing the approach moves the mean far more than the cool-down or the power cycle.

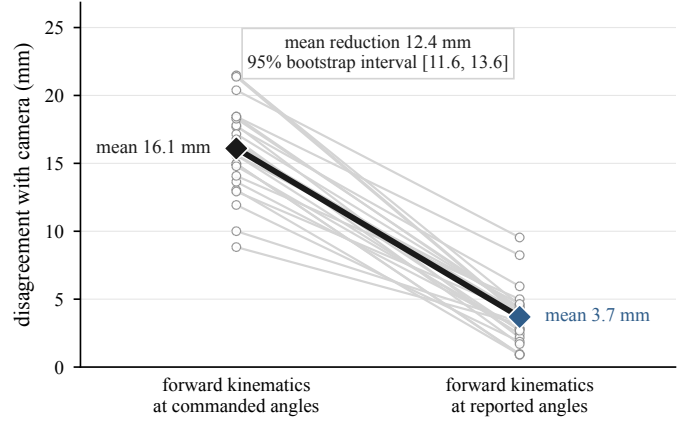


Fig. 3. Camera disagreement at each measured workspace pose, with forward kinematics evaluated at the commanded and at the reported angles. The mean reduction and its 95% bootstrap interval over the measured poses come from 2000 resamples with the registration refitted in each.

eight that failed did so because the board left the camera view. For each measured pose we compared the measured camera position with forward kinematics evaluated twice, once at the commanded angles and once at the reported ones.

Reported angles removed about three quarters of the mean disagreement (Fig. 3), from 16 to 4 mm. They help a great deal. What they leave is still several times the repeatability of Section IV, and it is error the servos cannot see. The positions nominal forward kinematics gives from the commanded and from the reported angles differ by 15.4 mm on average. That difference is a separate vector from the camera disagreement, so it is not the amount removed from the total.

Two caveats apply to that result. Refitting the registration

without the pose being scored, either one pose at a time or in disjoint folds, made the reported-angle residual slightly larger, so the values plotted here are the favourable ones for the encoder-based estimate. The eight poses that returned no reading are a source of selection bias, because visibility depended on where the arm settled.

The sign correction was found after this survey was recorded, so the numbers above are a reanalysis. We therefore repeated the survey seven weeks later with the sign set beforehand. That survey drew 33 poses around the board-visible poses found by a fresh scan (Section VI), each commanded from home, and 22 returned a reading. The mean disagreement was 15.7 mm at commanded angles and 3.7 mm at reported angles.

We also left the command path as the platform ships it. Servo gains were not retuned, and no loop re-commanded the arm until its reported angles matched the target. Either change would shrink the command-to-report difference. Neither would remove the disagreement that remains between the reported angles and the camera.

VI. A LOCAL MODEL CORRECTS LOCAL TARGETS

The remaining error is large enough to be worth modelling. Write q for the angles the servos report, c for a command, and c_{-1} for the command before it. The state estimate places the arm where it now stands,

$$\hat{p}_{\text{state}}(q) = p_{\text{fk}}(q; \hat{\theta}), \quad (1)$$

while the command prediction asks where a command will put it,

$$\hat{p}(c, c_{-1}) = p_{\text{fk}}(c; \hat{\theta}) + r(c, c_{-1}). \quad (2)$$

The analytic term p_{fk} has parameters $\hat{\theta}$ covering the base transform, a scale per link, a zero offset per joint, and the camera offset, nineteen in all. Four of them cannot be separated from the others by a measurement of camera position, so their values are not reported. The residual r is a ridge regression on the command, on trigonometric and cumulative pitch features, and on the sign of the last motion in each joint. It absorbs what a rigid model cannot express.

The model was fitted in the 23 September session. A 30-pose scan with the camera pointed down at the table found 13 poses from which the board was visible. The calibration run drew 100 poses around them, with a uniform jitter of up to 0.12 rad on the base and pitch joints and 0.2 rad on the wrist roll, and visited them one after another without returning home, so that the direction term of the residual saw both directions of approach. The board was seen at 77 of them.

The two expressions answer two different questions. Evaluated at the reported angles, Eq. (1) says where the arm is, and it does that well: its cross-validated residual on the calibration poses is 2.1 mm, against 4.5 mm for the nominal chain with only its alignment fitted. Eq. (2) asks instead where the arm will settle given a command, which is harder. Nominal kinematics leaves 20.9 mm there. The analytic term alone

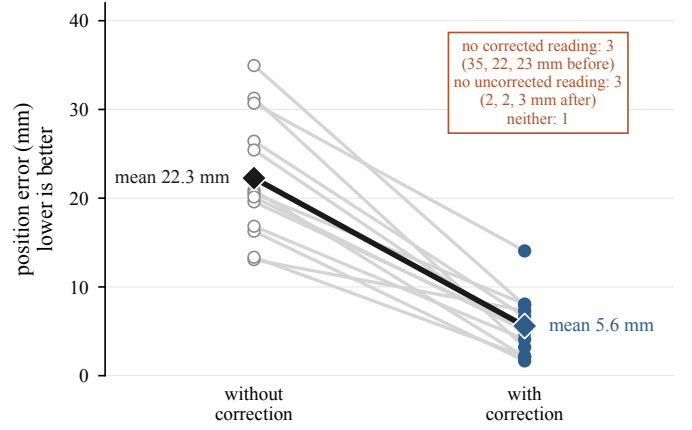


Fig. 4. Held-out targets approached once without and once with the correction, scored as the distance from the specified aim point. Twenty targets in two batches; the 13 measured in both passes are drawn, and the box lists the targets that lost the board in one pass or both.

leaves 20.2 mm, and adding the learned residual brings that down to 4.5 mm under fivefold cross-validation.

Correcting a target is a short fixed-point solve. It starts from the joint command that generated the target and runs two sweeps. Each sweep predicts the residual for the current command, given the command executed before it, subtracts that prediction from the desired position, and solves for the shifted point by damped least-squares inverse kinematics seeded from the current command, to a tolerance of 0.3 mm. A clamp holds the predicted residual to 50 mm as a guard against a corrupt model. The largest predicted residual was 32 mm, so the clamp never fired, and no joint moved by more than 6 degrees.

The targets were 20 poses drawn the same way as the calibration poses, around the same scan, in two batches of ten with seeds the fit never used. Each target was specified as a point: where the nominal chain, registered to the board on the calibration poses, places the camera for that pose. The fitted chain does not share a base frame with the nominal one, so the point is carried between them through the board. The uncorrected pass sent the target pose itself, and the corrected pass sent the command the model solved for the point. In each batch the uncorrected pass ran first. Both passes started from the same home position, visited the targets in the same order, and moved from target to target without returning home.

Thirteen targets returned a reading in both passes, and every one of them improved (Fig. 4). The mean distance from the aim point fell from 22.3 to 5.6 mm and the median from 20.5 to 5.1 mm. The worst corrected target missed by 14 mm. The paired reduction was 16.7 mm, with a 95% bootstrap interval over targets of 13.6 to 20.0 mm. The uncorrected misses are larger than the 16 mm of the survey because these targets were approached from the previous target rather than from home. The pose-to-pose calibration run gave a similar 20 mm at commanded angles.

Seven targets have no pair. Three lost the board only in the

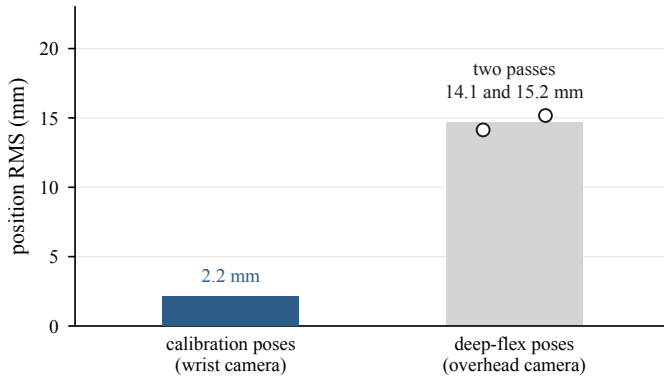


Fig. 5. The analytic model on the poses it was fitted to, and the same parameters held fixed on the deeply flexed set, where each pass keeps its own constant jaw offset.

corrected pass, three only in the uncorrected pass, and one in both, so each pass measured 16 of the 20 and the losses did not fall on the corrected pass. As a bound, if the correction had done nothing on the three targets it left unmeasured, the mean over the 16 targets with an uncorrected reading would still fall from 23.1 to 9.6 mm.

An earlier run of this experiment, on 7 August, computed its aim points with the roll sign as recorded, which moved each point by 14 to 34 mm. Its uncorrected baseline therefore mixed the arm's error with the sign error. The run reported here replaces it.

VII. THE FIT DOES NOT TRANSFER TO DEEP FLEXION

A fit that close at reported angles invites the conclusion that the model has recovered the arm's geometry. We tested that conclusion on 24 August, seventeen days after the first calibration session on 7 August. The analytic term was fitted again, on a fresh set of elevated and moderately flexed poses measured by the wrist camera. Its parameters were then held fixed and evaluated on a lower, deeply flexed set (Fig. 6), measured differently: an overhead camera tracked a cube held in the jaws, and each pass was allowed its own best constant offset from wrist to cube. The test covers only the analytic reported-angle term, not the learned command residual.

The fresh fit was accurate on its own poses. Held fixed, the same parameters left more than a centimetre of RMS error on the deeply flexed set, in both passes (Fig. 5). The nominal chain, given the same per-pass offset, left 11 and 13 mm there, so outside its own family the fit had not improved on the model it started from. The parameters of the 7 August calibration gave almost the same deep-flex values, so the age of the fit is not the explanation. Parameters that fit one family closely do not describe the other, so they cannot be read as a recovered geometry of the arm.

Registration and kinematics turn out to be one coupled estimate rather than two independent ones. Swapping the board-to-base registrations between the 7 August session and the re-fit session made the errors worse, even though an independent check found the board had moved by less than

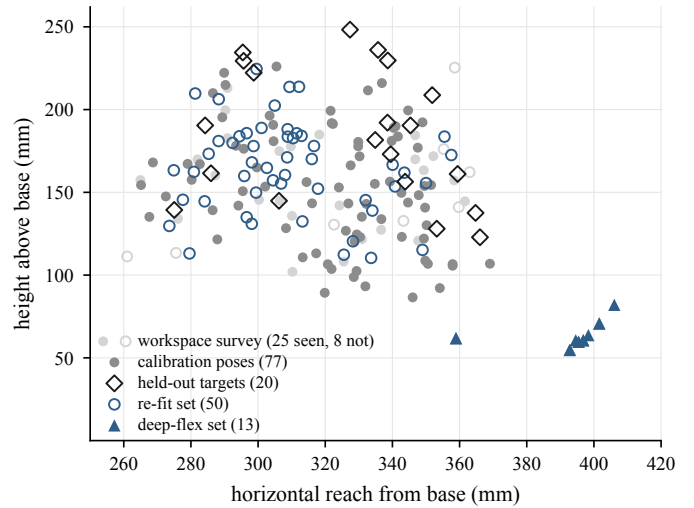


Fig. 6. Where the poses are. Tool position from nominal forward kinematics at the reported angles, in the base frame, for every pose set in the paper: the workspace survey of Section V, the calibration poses and held-out targets of Section VI (23 September), and the re-fit and deep-flex sets of this section (24 August).

a millimetre. Whatever the fit calls geometry includes some frame error. Because the base transform, the camera offset, the joint zero offsets, and the link scales can trade off against one another in explaining the same displacement, the fitted values should not be read individually as estimates of physical dimensions or encoder zeros. The sign check of Section III is the plainest example: with the roll sign as recorded, the same least-squares fit reached the same residual by folding the wrist back through a zero offset of 177 degrees and a link of negative length. Corrected, it reports a shoulder offset scaled by 0.6 and zero offsets of several degrees, and predicts the calibration poses exactly as well. Fitted again on 23 September, on poses of the same family, it returned zero offsets within 2 degrees of those values and a camera offset within 2 mm, but a shoulder offset scaled by 0.52. Within one family the fit repeats, which still does not make it the geometry of the arm.

Fitted per-joint parameters drift within a day as well. A simpler model of the servos alone, which maps each commanded angle to the angle reported back through a gain, an offset, and a backlash width, was fitted four times over one working day. Its offset for the gravity-loaded shoulder pitch moved by about 12 mrad across the four fits and that for the elbow by about 6 mrad, while the other three joints stayed within 1.5 mrad. These too are effective parameters of a condition, not constants of the mechanism.

VIII. LIMITATIONS

The following were not controlled.

- One arm, one set of printed parts, one assembly. Nothing here separates the behaviour of this unit from unit-to-unit variation.
- Missing readings are driven by visibility, and visibility depends on where the arm settled, so the poses and targets that went unmeasured are not missing at random.

- The wrist camera both fitted the correction and scored it. An independent instrument would be a stronger test.
- The calibration poses and the targets were drawn around poses from which the wrist camera sees the board, so the correction was tested only where the board is visible, inside the family it was fitted on.
- The nominal geometry is the published SO-ARM100 model. Part of what the fit reports as a link scale may be the difference between that model and the SO-ARM101 measured here.
- Only the translation of a single point was measured. Orientation and gripper contact were not.
- The transfer test changed the measured point and the camera together, so its values diagnose transfer rather than provide an instrument-matched comparison with the calibration poses.
- Nothing was tested at the level of a policy. Whether the correction improves learned control or task success is unknown.
- No outside replication. One person collected and analysed every measurement on one bench.

IX. CONCLUSION

On this arm a single command reaches two different places depending on the direction of approach. Reading the servos back recovers most of the disagreement with an external camera, and a local fit of the chain most of what remains, but neither makes the command itself accurate. A local model of the command-to-position map corrects targets inside the family of poses it was fitted on, and it corrects them well, from 22 to 6 mm on held-out targets. Its analytic parameters nonetheless fail on a differently loaded family, which means they encode an operating condition and not the geometry of the machine. The practical advice this paper offers is to treat the command, the reported state, and the physical state as three separate quantities. Accuracy claims for arms like this one should name the conditions under which they were measured.

CODE AVAILABILITY

The measurement and modelling code is released as vBody (Virtual Body) at <https://github.com/vladmarian20005/vbody>.

It is a command-line tool for the SO-ARM101 with no dependencies beyond NumPy, SciPy, OpenCV, and MuJoCo, the last used only to parse the arm model. It prints the ChArUco board, fits the wrist camera's intrinsics, scans for poses from which the camera sees the board, and runs the workspace survey of Section V around them. Every survey reports the disagreement at commanded and at reported angles and runs the joint-sign check of Section III. From a survey it fits the model of Section VI, the chain and board transform on reported angles and the residual on commands, with the same fivefold cross-validation, and it refuses to fit a survey whose sign check fails. It evaluates a held model on a later survey with the board placed anew, which is the test of Section VII. It approaches fresh targets with and without the correction and refuses any corrected command that would leave the joint limits or the table clearance. The repeat survey and the correction experiment of Section VI were run with it. A simulated arm with a deliberate geometry error, backlash, and sag runs the whole sequence without hardware.

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